A Position-Temporal Awareness Transformer for Remote Sensing Change Detection

Yikun Liu, Kuikui Wang, Mingsong Li, Yuwen Huang, and Gongping Yang

***Abstract*— With the development of deep learning, significant progress has been made in change detection (CD) methods for remote sensing (RS) images. However, many convolutional neural network (CNN)-based methods are constrained in capturing long-range dependencies due to the limitations of the receptive field. Transformers rely on self-attention mechanisms to effectively achieve global information modeling and are widely used in CD tasks. Nevertheless, transformer-based CD methods still suffer from issues such as pseudochanges and incomplete edges due to the lack of position and temporal correlations in bitemporal RS images. To deal with this issue, we propose a position-temporal awareness transformer (PT-Former), which models position and temporal relations in bitemporal images. Specifically, a Siamese network attached to a position-aware embedding module (PEM) serves as a feature encoder to extract the features of changed areas. Then, a temporal difference perception module (TDPM) is designed to capture the cross-temporal shift and enhance the difference perception ability during cross-temporal interaction. Meanwhile, the contextual information of the ground object is aggregated by the fusion block, and the spatial relation is reconstructed under the guidance of bitemporal features. The experimental results validate the superiority of PT-Former on three benchmark datasets, including the season-varying CD (SVCD) dataset, the learning vision and RS laboratory building CD (LEVIR-CD) dataset, and the WHU-CD dataset confirming the potential of PT-Former for CD tasks in RS images. The code will be available at https://github.com/liuyk29/PT-Former.**

***Index Terms*— Change detection (CD), feature fusion, position and temporal modeling, remote sensing (RS), transformer.**

I. Introduction

REMOTE sensing (RS) images change detection (CD) using image analysis techniques aims to compare and analyze the surface changes between coregistered pairs of images captured at different times in the same regions [1]. It is an important task in RS images analysis and has been widely used in damage assessment [2], urban planning [3], land use monitoring [4], and building CD [5]. In recent years, the role of CD in land use and urban planning has become more prominent due to the advancement of society and the economy. Moreover, the progress in RS technology has enabled the acquisition of high-resolution multitemporal RS images, thereby significantly advancing the field of CD [6].

Nevertheless, CD still presents various challenges. Initially, bitemporal RS images are taken under distinct imaging conditions and at different time gaps, resulting in color or shape discrepancies in ground objects caused by seasonal variations, lighting conditions, and physical transformations. In addition, high-resolution multitemporal RS images encompass intricate details of ground objects, requiring CD models to have sturdy feature recognition abilities. These obstacles complicate the process of extracting semantic features in temporal and spatial aspects. Hence, a proficient CD model should demonstrate robust cross-temporal interaction and semantic extraction capabilities.

In recent years, convolutional neural networks (CNNs) have been extensively used in various RS tasks, such as scene classification [7], [8], [9], object detection [10], [11], semantic segmentation [12], and CD [13], [14], [15]. Numerous CNN-based approaches have been developed to enhance the accuracy of CD. For instance, a method employing an edge-guided recurrent CNN incorporated edge structure details for CD improvement [16]. In addition, generative adversarial training was utilized to produce bitemporal images with diverse changes to tackle data scarcity issues [17]. Furthermore, an unsupervised progressive learning framework was introduced to eliminate inaccurate change information using a label selection filter [18]. Ultimately, a spatiotemporal enhancement and interlevel fusion network is proposed to improve the ability of feature representation for changing objects [19].

Recently, the proposed transformer has demonstrated significant promise in modeling long-range dependencies within images and has been utilized across a range of tasks, such as object detection [20], [21], semantic segmentation [22], and CD [23], [24]. By jointly considering spatiotemporal dependencies, a semantic change transformer (SCanFormer) is proposed to explicitly model the semantic transitions between bitemporal RS images [25]. To pay attention to the edges of changed areas, an edge-assisted adaptive transformer detector is proposed, which refines the change and edge detection results [26]. To obtain an effective change feature, an attention-filtering network based on a branch transformer is used to efficiently fuse features of a different temporal and perform

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progressive information filtering on the extracted multilevel features [27]. To find small changes and extract more accurate edge information, a bidirectional interactive framework based on Transformer is proposed with an asymmetric interaction attention module [28].

Although many transformer-based CD methods have been proposed, there are still some issues present in these methods. First, these transformer-based methods merely regarded the transformer as a fundamental feature extractor. They overlook incorporating ground object position details from the feature map, which leads to the fact that some pixels at the edges of the building are the most likely to be missed when detecting a changing building. Second, while these methods have delivered satisfactory results, it is essential to carefully consider the incorporation of cross-temporal modeling through dual-stream features. This limitation impedes the accurate distinguishing between bitemporal images, consequently impacting the detection results.

To address these challenges, this article proposes a position-temporal awareness transformer called PT-Former, which achieves accurate temporal identification and sensitivity to change position. We believe that modeling the position and temporal relation in bitemporal images can help precisely capture the changing areas and cross-temporal differences (CTDs). In order to model the position relation of features more accurately, we simultaneously extract local and global information to build position-aware embeddings. To model the cross-temporal correlations of features, we define a learnable temporal embedding to characterize the CTDs. In addition, we consider that simply merging these features may disregard the possible relationships among them. We fuse and separate the bitemporal features at multiple stages, achieving full cross-temporal interaction of bitemporal features.

In summary, the main contributions of this work are given as follows.

1) To enhance the capacity for perceiving spatial changes, we design a position-aware embedding module (PEM) within the encoder, aiming at modeling the positional information inherent in bitemporal images.
2) We propose a temporal difference perception module (TDPM) to capture cross-temporal shifts and enhance the discernment of changes during temporal interactions.
3) We design the feature fusion blocks to integrate contextual information of ground objects at multiple stages while reconstructing spatial relations guided by bitemporal features.

This article is organized as follows. In Section II, the related work is introduced. The proposed PT-Former is described in detail in Section III. The settings of all experiments are presented in Section IV. Section V presents and analyzes the experimental results. Section VI concludes this article.

II. Related Works

A. Traditional CD Methods

Tradition CD methods can be categorized into two categories: Pixel-based CD and object-based CD [29]. Pixel-based CD methods usually calculate the difference map by directly computing the spectral or texture feature of the bitemporal RS images and generating the change map by a suitable threshold or clustering. For mapping forest cover change, a training data automation procedure and support vector machine (SVM) algorithm is proposed [30]. In order to achieve high CD accuracy, extreme learning machine (ELM), multinomial logistic regression (MLR), and K-nearest neighbor (KNN) classifiers were exploited in combination [31]. Principal component analysis (PCA) and k-means clustering were used for unsupervised CD in multitemporal satellite images [32]. To compensate for the omission of spatial context information by the pixel-based CD methods, object-based CD methods are studied, which divide bitemporal RS images into many regions based on the similarity of spectral and spatial and extract features from these regions to identify changes [33]. A nonlocal patch similarity-based graph was built to measure the structure consistency between heterogeneous RS images [34]. A multilevel object-based change vector analysis (CVA) approach was proposed to integrate the change information in multiple segmentation scales [35]. In addition, various priors of changed information, such as sparsity [36], low rank [37], and multiscale property [38], in the observed scenes were also explored for CD. However, these abovementioned methods rely too much on manually extracted features and cannot achieve satisfactory results on high-resolution RS images, which have more details of ground objects.

B. CNN-Based CD Methods

Currently, deep learning has advanced as an important technique for CD due to its ability to generate precise CD predictions. In contrast to traditional approaches, CNN-based methods can systematically extract spatial, spectral, and temporal features of bitemporal images in feature space. The features learned by CNNs possess more robust representation capabilities compared to manually crafted features from traditional methods. Consequently, CNN-based methods attain superior CD accuracy, and several methods have been proposed to improve CD accuracy. For example, an edge-guided recurrent CNN utilized edge structure information for CD [16]. Generative adversarial training was employed to generate bitemporal images containing diverse changes, addressing data insufficiency [17]. An unsupervised progressive learning framework filtered out incorrect change information from the region using a label selection filter [18]. Parallel convolutional structures and self-attention were used to enhance feature extraction and reduce computational cost [39]. A boundary extraction-constrained Siamese network was proposed to leverage boundary information [40]. Attention and transformer-based networks were utilized to improve feature representation [41]. Transformer-based methods combined with U-Net were used to capture global context information [42]. Selective attention modules were developed to model the relationship between semantics and pixel state, integrated into metric learning-based architectures [43]. A pair-to-video CD framework treated CD as a video understanding problem by constructing a pseudotransition video [44]. Generative

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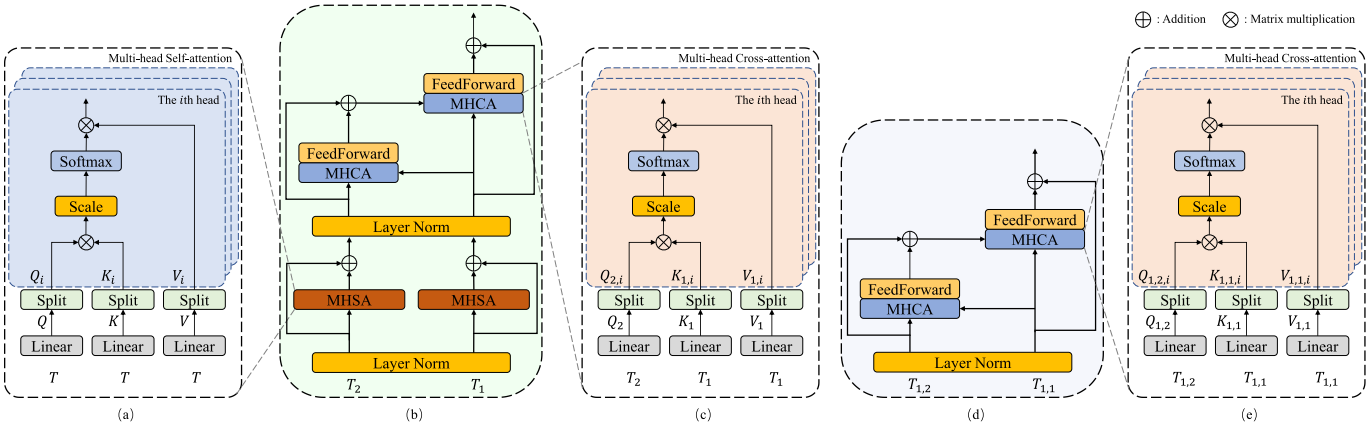


Fig. 1. Architecture of (a) multihead self-attention, (b) transformer cross-encoder, (c) multihead cross-attention of the encoder, (d) transformer cross-decoder, and (e) multihead cross-attention of the decoder.

adversarial networks were employed to learn superresolution images and overcome resolution differences between bitemporal images [45]. The self-attention module was used to calculate the attention weights between any two pixels at different times and positions and uses them to generate more discriminative features [46]. SNUNet-CD alleviates the loss of localization information in the deep layers of the neural network through compact information transmission between encoder and decoder, and between decoder and decoder [47]. To improve the ability to detect changes in complex scenes, feature interaction and multitask learning model the context information of features through a multilevel feature interaction module and improve the sensitivity of the model to changes [48].

Recently, there has been a growing focus among researchers on the attention mechanism. Jiang et al. [49] proposed an efficient multidimensional attention-aggregation network, which keeps better feature aggregation while maintaining excellent differential attention ability. Li et al. [50] proposed a lightweight network, which identifies changes based on the features extracted by mobile networks via progressive feature aggregation and supervised attention. Zhang et al. [51] proposed a multiscale cascaded cross-attention hierarchical network, which uses a large kernel convolution formed by stacking small kernel convolutions combined with an efficient transformer as the backbone network to achieve local and global feature extraction and fusion. Li et al. [52] proposed spatial–temporal attention with a difference enhancement-based network, and a CD difference enhancement module is proposed to extract important features from the difference map to detect changed regions. Many CNN-based methods utilize the Siamese networks. Nevertheless, these methods have primarily focused on identifying basic land cover changes; it is necessary to enhance their robustness to handle intricate edge details and small targets [53].

C. Transformer-Based CD Methods

The transformer proposed by Vaswani et al. [54] was initially applied in natural language processing (NLP) tasks [55], which demonstrates outstanding performance and attracting significant interest. Likewise, the enhanced vision transformer (ViT) [56] has significantly contributed to various computer vision tasks, such as CD, and has emerged as a prominent trend. ViT divides the input image into patches of equal size, converting each patch into a standardized token that is then fed into the multihead attention mechanism. As shown in Fig. 1(a), within the multihead attention process, the attention weight of a Query (Q), which represents the image's prompt, is computed. The key (K) and the corresponding value (V) are multiplied together. Essentially, the calculation determines the degree of attention or relationship between Q and K, followed by the multiplication of the corresponding V linked to each K. Chen et al. [23] represented the high-level concepts of the change of interest by semantic tokens and used a transformer encoder to model contexts. ChengeFormer [57] unified hierarchically structured transformer encoder with multilayer perception (MLP) decoder in a Siamese network architecture to efficiently render multiscale long-range details required for accurate CD. Zhang et al. [58] introduced a U-shaped Swin-transformer-based network to improve the model's global exploration capabilities. This Siamese network employs a multilevel Swin-transformer block to construct an encoder and decoder framework, enhancing efficient CD. Wang et al. [59] combined the advantages of a multiscale transformer with the convolutional block attention module (CBAM) to improve the detection quality of different RS images. Zhang et al. [24] analyzed the relation changes in multitemporal images and proposed a CTD attention to capturing these changes efficiently.

These transformer methods for feature extraction do not fully utilize the power of transformers. They define cross-temporal feature interactions as temporal extraction or modeling, which is not advanced in terms of modeling temporal and positional information. PT-Former uses a transformer-based embedding for temporal and positional modeling, which is more scientific and effective.

III. Method

In this section, we first introduce the overall structure of our model. PT-Former mainly includes four modules, which are

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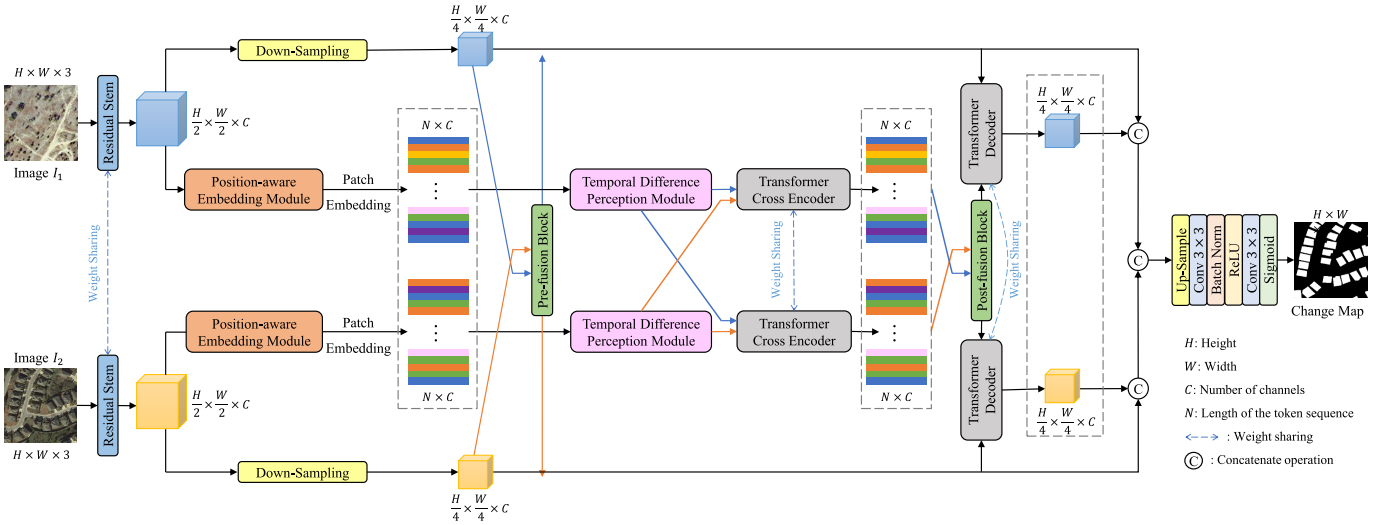


Fig. 2. Overall architecture of the proposed PT-Former.

the transformer, PEM, TDPM, and the fusion block. They will be introduced in detail in the following. In addition, we also give the loss function of our model.

A. Overview

The overview structure of the PT-Former is shown in Fig. 2. PT-Former takes three channels of bitemporal RS images as input and outputs two channels of change prediction maps. The bitemporal RS images input to PT-Former are progressively fed into a dual-branch residual stem with shared weights. In the proposed PT-Former, a modified ResNet-18 [24] is employed for feature extraction. After the spatial resolution has been reduced and the semantics have been extracted, the feature maps are fed into the PEM to generate position embedding, and the features are embedded by PEM and a flattened operation. To capture the cross-temporal shift and enhance the difference perception ability during cross-temporal interaction, the bitemporal features are fed into TDPM. The feature continues through the transformer cross-encoder with cross-attention, and the features are reshaped. Simultaneously, the feature maps from the residual stem are downsampled and fused by the pre-fusion block. Next, the features extracted from the transformer encoder are aggregated by the post-fusion block. Then, the fused features are fed into the transformer decoder.

In addition, to enhance the features of changed areas, the features extracted by the transformer decoder and the post-fusion block are concatenated. Finally, to infer the change map of the observed scene, a classifier head is introduced to generate the change map.

B. Transformer Block

1) Revisiting the Self-Attention in Transformer: In the transformer-based vision task, the feature map F with the size of H × W × C is usually flattened as a token T with the size of $N \times$ C as the input of the transformer block, where H donates the height of the feature map, W donates the width of the feature map, C donates the number of channels of the feature map or tokens, and $N = \mathrm{HW}$ donates the length of the token sequence. In the transformer block, the token T projected to the query Q, K, and V embedding. The self-attention of the transformer is calculated by the following formula:

$$\text{Self-Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (1)$$

where d is a scale factor.

2) Transformer With Cross-Attention: In order to enhance the cross-temporal features interaction, we introduce the transformer cross-encoder with cross-attention. Specifically, we receive the bitemporal tokens T_1 and T_2 as inputs and project them to Q_1, K_1, V_1 and Q_2, K_2, V_2, respectively. We perform cross-attention on bitemporal embeddings. The cross-attention is calculated as follows:

$$\text{Cross-Attention}_1(Q_2, K_1, V_1) = \text{softmax}\left(\frac{Q_2K_1^T}{\sqrt{d}}\right)V_1 \quad (2)$$

$$\text{Cross-Attention}_2(Q_1, K_2, V_2) = \text{softmax}\left(\frac{Q_1K_2^T}{\sqrt{d}}\right)V_2. \quad (3)$$

According to the cross-attention, we use the transformer cross-encoder to further extract the change information from bitemporal tokens T_1 and T_2. Fig. 1(b) shows the architecture of the transformer cross-encoder for the token T_1. The transformer cross-encoder is composed of two multihead self-attention and two multihead cross-attention. Thus, the process in Fig. 1(b) can be formulated as follows:

$$z_1 = \text{LN}(T_1) + \text{MHSA}(\text{LN}(T_1)) \quad (4)$$

$$T_2 = \text{LN}(z_2) + \text{FF}(\text{MHCA}(\text{LN}(z_2), \text{LN}(z_1))) \quad (5)$$

$$T_1 = \text{LN}(z_1) + \text{FF}(\text{MHCA}(T_2, \text{LN}(z_1))) \quad (6)$$

where $\text{LN}(\cdot)$ is the layer normalization, $\text{MHSA}(\cdot)$ is the multihead self-attention, $\text{MHCA}(\cdot)$ is the multihead cross-attention, and $\text{FF}(\cdot)$ is the feedforward layer.

Within PT-Former, the multihead mechanisms depicted in Fig. 1(a) and (c) are designed to leverage representations

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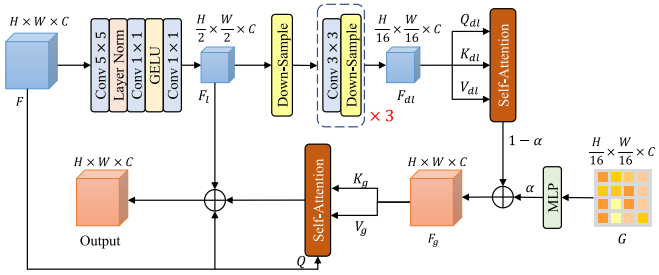


Fig. 3. Architecture of PEM.

across various subspaces. Utilizing multiple heads allows for a more effective exploration of attention patterns among query, key, and value components. To simplify the computational complexity, the projected features are divided into N parts along the feature dimension; N indicates the number of heads in Fig. 1(a) and (c). Finally, the multihead self-attention MHSA$(\cdot)$ and multihead cross-attention MHCA$(\cdot)$ are represented as follows:

$$\text{MHSA}(\text{LN}(T_1)) = \text{Concat}(\text{shead}_1, \ldots, \text{shead}_N) \quad (7)$$

$$\text{MHCA}(\text{LN}(z_2), \text{LN}(z_1)) = \text{Concat}(\text{chead}_1, \ldots, \text{chead}_N) \quad (8)$$

where shead_i and chead_i are indicate the attention in (1) and (2)

$$\text{shead}_\text{i} = \text{Self-Attention}(Q, K, V) \quad (9)$$

$$\text{chead}_\text{i} = \text{Cross-Attention}(Q_2, K_1, V_1). \quad (10)$$

The token T_2 is processed with the same architecture, as shown in Fig. 1(b).

To pass the change information from the fusion blocks, the transformer decoders with cross-attention are constructed. Fig. 1(d) shows the architecture of the transformer decoder for the tokens sourced from the fusion blocks. For the image of pretemporal, the tokens from the two fusion blocks are $T_{1,1}$ and $T_{1,2}$, respectively. The keys $K_{1,1}$ and $V_{1,1}$ with the size of $N \times \text{C}$ are projected by the linear projection from $T_{1,1}$. Then, the query $Q_{1,2}$ with the size of $N \times \text{C}$ is obtained from $T_{1,2}$. Thus, the process of transformer cross-decoder is represented as

$$z_{1,2} = \text{FF}\big(\text{MHCA}\big(\text{LN}\big(T_{1,2}\big), \text{LN}\big(T_{1,1}\big)\big)\big) \quad (11)$$

$$T_1 = \text{FF}\big(\text{MHCA}\big(\text{LN}\big(T_{1,1}\big), z_{1,2}\big)\big). \quad (12)$$

In the transformer cross-decoder, the projected features are split into R parts along the horizontal dimension to reduce the complexity. Fig. 1(e) shows the multihead cross-attention of the decoder; R is the number of heads. The multihead cross-attention MHCA$(\cdot)$ is represented as follows:

$$\text{MHCA}\big(\text{LN}\big(T_{1,1}\big), z_{1,2}\big) = \text{Concat}(\text{chead}_1, \ldots, \text{chead}_R). \quad (13)$$

Similarly, T_2 is also obtained with the transformer cross-decoder to refine the change information in the post-temporal image.

Finally, we obtain the enhanced change features at fine scales, which are then concatenated with other features.

C. Position-Aware Embedding Module

Currently, numerous transformer-based methods exist for capturing the long-range dependencies. However, these methods do not fully harness the capabilities of transformers, as they rely on spatial attention mechanisms to capture spatial information. In addition, for the CD task, the positional relationship between features is extremely important; for example, when detecting changing buildings in the learning vision and RS laboratory building CD (LEVIR-CD) dataset, some pixels at the edges of the building are the most likely to be missed, and with the introduction of positional embedding, the positional information of the edge pixels makes them easier to be detected. Toward this end, PEM is designed to learn both global change-related information and fine-grained local context between bitemporal feature maps.

Generally, normal transformer tokens are 1-D, offering limited positional information. In contrast, our position embeddings are 2-D, thus encapsulating a richer array of location information. This expansion allows for more precise localization of changes, thereby enhancing the model's ability to perceive spatial changes. The architecture are shown in Fig. 3,

To enhance the capacity for perceiving spatial changes, we extract the local and global position information, respectively, and then fuse these two parts. First, we use a convolution block for local position information extraction in the PEM of our model since local position information extraction with convolution will perform better than window self-attention. The convolution block is represented as follows:

$$F_l = F + \text{PW}_1(\text{GELU}(\text{PW}_2(\text{LN}(\text{DW}(F))))) \quad (14)$$

where F is the input feature maps of size H × W × C, DW is the depthwise convolution with a kernel size of 5, $\text{PW}_1(\cdot)$ and $\text{PW}_2(\cdot)$ are pointwise convolution with a kernel size of 1, and F_l is the output with local position information.

In order to reduce the complexity of extracting global position information, the local position information F_l is downsampled and aggregated with the global position information. The downsampling process is repeated three times until the feature map size reaches the expected size (H/16) × (W/16) × C. Compared with the one-step downsample method, the stepwise downsample method can reduce the loss of semantic information and retain more useful information. The entire downsample process is represented as follows:

$$F_{dl} = \{\text{DS}(\text{PW}(F_l))\}_3 \quad (15)$$

where $\text{PW}(\cdot)$ represents the convolution with kernel size 3 and $\text{DS}(\cdot)$ represents the downsample operation.

The global position information F_g is aggregated with self-attention, and we use learnable position tokens to enrich global position information F_g. Before fusion, we employ MLP for the learnable position tokens. The process is represented as follows:

$$F_g = (1-\alpha)\text{SA}(Q_{dl}, K_{dl}, V_{dl}) + \alpha\text{MLP}(G) \quad (16)$$

where Q_{dl}, K_{dl}, and V_{dl} are produced by F_{dl} through linear projection. $\alpha \in [0, 1]$ is a weight set in advance. $\text{SA}(\cdot)$ indicates the self-attention. $\text{MLP}(\cdot)$ indicates the MLP layer.

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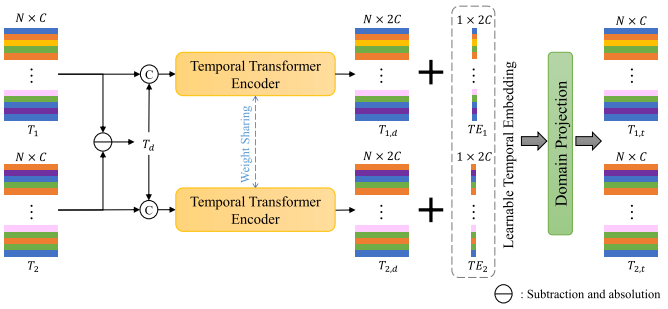


Fig. 4. Architecture of TDPM.

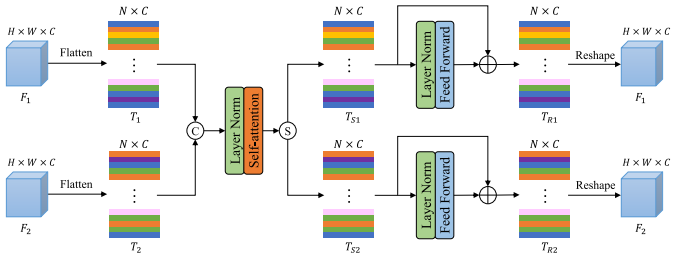


Fig. 5. Architecture of the fusion block.

The global position information is broadcast to the input feature maps using self-attention. Finally, the output feature maps contain local and global position information, respectively,

$$\text{Output} = F_l + \text{SA}\big(Q, K_g, V_g\big) + F \tag{17}$$

where Q is produced by the input feature maps through linear projection, and K_g and V_g are produced by F_g through linear projection.

D. Temporal Difference Perception Module

Another issue of bitemporal image CD is the pseudochanges, which are mainly caused by temporal-related factors such as different acquisition times, lighting conditions, and seasonal changes. For example, the same building captured during cloudy and sunny days also has large visual and feature differences, which is defined as the cross-temporal shift in this article, and the cross-temporal shift between the pretemporal images and the posttemporal images of the whole dataset tends to be the same. To address this issue, we attempt to capture this cross-temporal shift and enhance the temporal discrimination ability when the features interact by incorporating the learnable temporal embedding into the features of the pretemporal and posttemporal and ultimately eliminate the cross-temporal shift. The architecture is shown in Fig. 4.

Through the patch embedding in Fig. 2, the bitemporal features are embedded and flattened in tokens T_1 and T_2 with the shape of $N \times \text{C}$. In TDPM, T_1 and T_2 are first subtracted to get the cross-feature difference token T_d with absolution operation. Then, we concatenate T_d with T_1 and T_2 to obtain the differential fusion feature. In order to extract temporal information from features, we fed the differential fusion feature into a temporal transformer encoder to obtain the features $T_{1,d}$ and $T_{2,d}$

$$T_d = |T_1 - T_2| \tag{18}$$

$$T_{1,d} = \text{TTE}(\text{Concat}(T_d, T_1)) \tag{19}$$

$$T_{2,d} = \text{TTE}(\text{Concat}(T_d, T_2)) \tag{20}$$

where $\text{TTE}(\cdot)$ indicates the temporal transformer encoder.

To reduce the pseudochanges caused by cross-temporal feature shifts, we have defined learnable temporal embeddings TE_1 and TE_2. The size of TE_1 and TE_2 is $1 \times 2C$. We add the corresponding temporal embeddings $TE1$ and TE_2 to $T_{1,d}$ and $T_{2,d}$, which eliminates the impact of cross-temporal shift on features $T_{1,d}$ and $T_{2,d}$. Finally, the temporal perception features $T_{1,t}$ and $T_{2,t}$ are projected by the linear projections

$$T_{1,t} = \text{DP}\big(TE_1 + T_{1,d}\big) \tag{21}$$

$$T_{2,t} = \text{DP}\big(TE_2 + T_{2,d}\big) \tag{22}$$

where $\text{DP}(\cdot)$ indicates the domain projection. The size of $T_{1,t}$ and $T_{2,t}$ is $N \times \text{C}$.

E. Fusion Block

The feature fusion of bitemporal features in CD plays a crucial role, which forms the complete change information under the guidance of bitemporal features. In addition, to integrate contextual information and reconstruct the spatial relations of ground objects, we design the feature fusion block to fuse bitemporal features at multiple stages, as shown in Fig. 5 (pre-fusion block and post-fusion block).

First, the bitemporal features F_1 and F_2 with the size of $\text{H} \times \text{W} \times \text{C}$ are flattened to the tokens T_1 and T_2 with the size of $N \times \text{C}$, where $N = \text{HW}$. Then, T_1 and T_2 are concatenated, and layer normalization is employed. To capture the relation changes of bitemporal tokens, self-attention is introduced to extract the global contextual information. The process can be written as follows:

$$T_1 = \text{Flatten}(X_1),\ T_2 = \text{Flatten}(X_2) \tag{23}$$

$$T_S = \text{SA}(\text{LN}(\text{Concat}(T_1, T_2))) \tag{24}$$

where $\text{Flatten}(\cdot)$ indicates the flatten operation. Next, the fused feature T_S is split into bitemporal tokens T_{S1} and T_{S2}. Then, layer normalization and a feedforward layer are employed to T_{S1} and T_{S2}, and a residual structure is introduced to simplify the model learning process

$$T_{S1}, T_{S2} = \text{S}(T_S) \tag{25}$$

$$T_{R1} = \text{FF}(\text{LN}(T_{S1})),\ T_{R2} = \text{FF}(\text{LN}(T_{S2})) \tag{26}$$

where $\text{S}(\cdot)$ indicates the split operation. The feedforward layer consists of two linear layers with the Gaussian error linear unit (GELU) activation function. Finally, the bitemporal features F_1 and F_2 are reshaped by T_{R1} and T_{R2}.

F. Loss Function

In RS CD, a prevalent issue that we encounter is the notable imbalance in class distribution between negative and positive pixels. This imbalance frequently impedes effective

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optimization, potentially causing the network to get stuck in local minima of the loss function. To tackle this challenge and improve learning from intricate scenes, we suggest employing a hybrid loss function that merges binary cross-entropy (BCE) loss with Dice loss. The formulation of the hybrid loss function is given as follows:

$$L_{\text{hybrid}} = L_{\text{BCE}} + \lambda L_{\text{Dice}} \tag{27}$$

where λ is the coefficient defined in the hybrid loss function.

To explain the concepts of BCE loss and Dice loss, we consider the predicted change map $\hat{Y}$ and the reference map Y as sets of pixels, represented, respectively, as $\hat{Y} = \{\hat{y}_i, i = 1, 2, \ldots N\}$ and $Y = \{y_i, i = 1, 2, \ldots N\}$. Here, $\hat{y}_i$ denotes the probability of change in the ith pixel, while y_i represents the reference value of the ith pixel. In this context, a value of 0 indicates an unchanged pixel, whereas a value of 1 signifies a changed pixel. The total count of pixels in the change map is denoted by N. The combined objective function of these two loss functions can be formulated as

$$L_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N} y_i \log_2 \hat{y}_i + (1 - y_i)\log_2\left(1 - \hat{y}_i\right) \tag{28}$$

$$L_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \hat{y}_i}. \tag{29}$$

IV. Experiment Setting

A. Datasets

In order to evaluate the performance and robustness of our PT-Former, we carried out comparative experiments on three challenging datasets, including the season-varying CD (SVCD) dataset [60], the LEVIR-CD dataset [46], and the WHU-CD building dataset (WHU-CD) [61].

1) SVCD: The SVCD dataset comprises 16 000 Google Earth image pairs with a size of 256 × 256 pixels, featuring varying seasons. The dataset consists of 10 000 training samples, 3000 validation samples, and 3000 testing samples, with a high spatial resolution of 3–100 cm. The SVCD dataset provides not only change information on common objects such as buildings, roads, and forests but also many detailed objects such as cars and tanks.

2) LEVIR-CD: The LEVIR-CD dataset consists of 637 image pairs, each with a size of 1024 × 1024 pixels and a spatial resolution of 0.5 m. Due to GPU memory limitations, the images are uniformly cut into 256 × 256 patches, resulting in 7120/1024/2048 pairs of patches for training, validation, and testing, respectively.

3) WHU-CD: The WHU-CD dataset is generated from one pair of aerial images captured from Christchurch, New Zealand. The spatial size of the dataset is 32 507 × 15 354, and the spatial resolution is 0.2 m. The dataset is also divided into small patches with size 256 × 256 for training, validation, and testing. The dataset is divided randomly into 5947/744/745 for training, validation, and testing, respectively.

B. Implementation Details and Model Design

The proposed model is implemented using the PyTorch framework and is trained on an NVIDIA Geforce RTX 3090 GPU with 24 GB of video memory. Common data augmentation techniques are applied to the input image patches. A batch size of 16 is used, and the initial learning rate is set to $5e-4$ for both three datasets. Stochastic gradient descent (SGD) with momentum is employed to optimize the model, and the momentum is set to 0.9, while the weight decay is set to 0.0002. The hybrid loss function coefficient is set to 1.4, 1, and 1.2 for the SVCD, LEVIR-CD, and WHU-CD datasets, respectively; the experiment results are shown in Section V-D4. The model is trained for 80 000 iterations.

C. Evaluation Metrics

To measure the performance of models in the experiments, we use five metrics: precision, recall, $F1$-score, IoU, and Kappa coefficient. $F1$-score, IoU, and Kappa coefficient act as comprehensive metrics that are indicative of general model performance, whereas high metric values represent positive model performance. The abovementioned metrics can be formulated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{30}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{31}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{32}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{33}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{34}$$

$$P = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN}) + (\text{FN} + \text{TN})(\text{TP} + \text{TN})}{(\text{TP} + \text{FP} + \text{TN} + \text{FN})^2} \tag{35}$$

$$\text{Kappa} = \frac{\text{OA} - P}{1 - P} \tag{36}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively. P denotes the hypothetical probability of chance agreement between the ground truth and predictions.

D. Comparative Methods

In this section, we make a comparison to several state-of-the-art methods, which are listed in the following.

1) FC-EF [62] utilized an early fusion method, employing a UNet-based network architecture. It concatenated original bitemporal images as network input, processing them through a single-stream convolutional network for results.
2) FC-Siam-Diff [62] implemented a feature-difference technique that extracts multilayer features of bitemporal images from Siamese networks for difference detection.
3) FC-Siam-Conc [62] followed a feature-concatenation method and extracted multilayer features of bitemporal images from Siamese networks for differential detection.
4) IFN [63] proposed a deeply supervised difference discrimination network for results. It fused multilevel deep

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TABLE I
QUANTITATIVE RESULTS OF ALL METHODS ON THE THREE CD DATASETS. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

Method	Params.(M)	FLOPs (G)	SVCD			LEVIR-CD			WHU-CD		
			Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
FC-EF	1.35	3.24	80.99	52.24	63.51	85.04	71.02	77.40	79.44	71.19	75.09
FC-Siam-Diff	1.35	4.38	87.51	44.37	58.89	88.98	84.71	86.79	49.27	79.29	60.77
FC-Siam-Conc	1.55	4.99	86.20	48.70	62.24	88.51	87.11	87.80	36.62	83.30	50.88
IFN	50.44	82.26	95.89	**97.91**	96.89	90.43	90.06	90.24	93.58	92.54	93.06
SNUNet	12.03	46.70	96.64	96.74	96.69	92.42	89.80	91.09	79.92	81.66	80.78
BIT	2.99	8.75	95.21	92.74	93.36	90.33	89.56	89.94	85.85	87.85	86.84
ChangeFormer	29.75	21.18	97.95	97.86	97.89	**93.04**	89.34	91.15	95.38	91.99	93.66
Changer	11.32	5.96	94.37	91.66	93.00	92.52	88.36	90.68	81.66	83.78	82.71
A2Net	3.78	3.05	97.46	96.59	97.03	92.90	90.59	91.73	96.04	91.91	93.93
SEIFNet	27.91	8.37	96.47	96.01	96.24	92.49	89.46	90.95	91.29	88.12	89.68
Ours	16.56	41.43	**98.45**	97.35	**97.90**	92.95	**91.04**	**91.99**	**96.09**	**92.63**	**94.33**

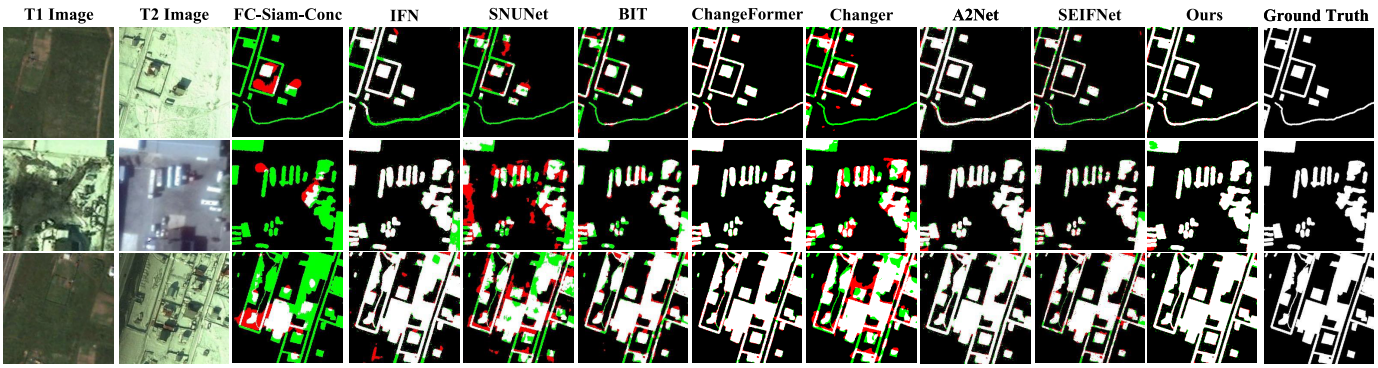


Fig. 6. Qualitative comparisons on the SVCD dataset. Different colors are used for a better view, i.e., white represents true positives, black represents true negatives, red represents false positives, and green represents false negatives. The less the green and red, the better the performance represented.

features with image difference features using attention modules for change map reconstruction.

5) SNUNet [47] designed a densely connected deep SNN with strong feature extraction capabilities. It introduced an ensemble CBAM to refine features.
6) BIT [23] introduced a bitemporal image transformer within a deep feature differencing-based framework to model contexts in the spatial–temporal domain.
7) ChangeFormer [57] extracted the bitemporal features by cascaded transformer blocks, and the difference features were decoded by a lightweight MLP.
8) Changer [64] introduced a new general results architecture incorporating two interaction strategies: aggregation-distribution and exchange. A comparison is made between its ChangerEX architecture and our PT-Former.
9) A2Net [50] proposed a novel lightweight network, which identifies changes based on the features extracted by mobile networks via progressive feature aggregation and supervised attention.
10) SEIFNet [19] introduced the spatiotemporal difference enhancement modules to capture global and local information from bitemporal feature maps at each level.

V. RESULTS AND DISCUSSION

A. Comparison to State-of-the-Art Methods

1) Results on the SVCD Dataset: The quantitative comparison results are shown in Table I. The best values are marked in bold. The quantitative comparison results show that the performance of the proposed PT-Former on SVCD datasets is significantly better than other comparative methods. The PT-Former achieves the highest values in terms of precision and $F1$ in the SVCD dataset. Compared to the second higher method ChangeFormer, PT-Former further improves the precision and $F1$ to 98.45% and 97.90%. The IFN obtains the best recall of 97.91%. The largest $F1$ value means that the CD results of PT-Former achieve the best accuracy and fewer pseudochanges. The performance of PT-Former may benefit from the position and temporal modeling.

Fig. 6 shows the CD results of all methods on different image pairs. In order to show the detection results more visually, we use different colors to indicate TP (white), TN (black), FP (red), and FN (green). PT-Former shows better performance in detecting shadow regions compared to other methods. Moreover, PT-Former can detect building changes more accurately in noisy environments. It is worth mentioning that the images from two different temporals in the SVCD dataset have significant distribution differences. Benefiting from TEM, PT-Former is able to learn the cross-temporal distribution differences and a certain degree of adjustments, thus enabling more accurate identification of the change regions.

2) Results on the LEVIR-CD Dataset: The quantitative comparison results are shown in Table I. The best values are marked in bold. The quantitative comparison results show that the performance of the proposed PT-Former on LEVIR-CD datasets is significantly better than other comparative methods.

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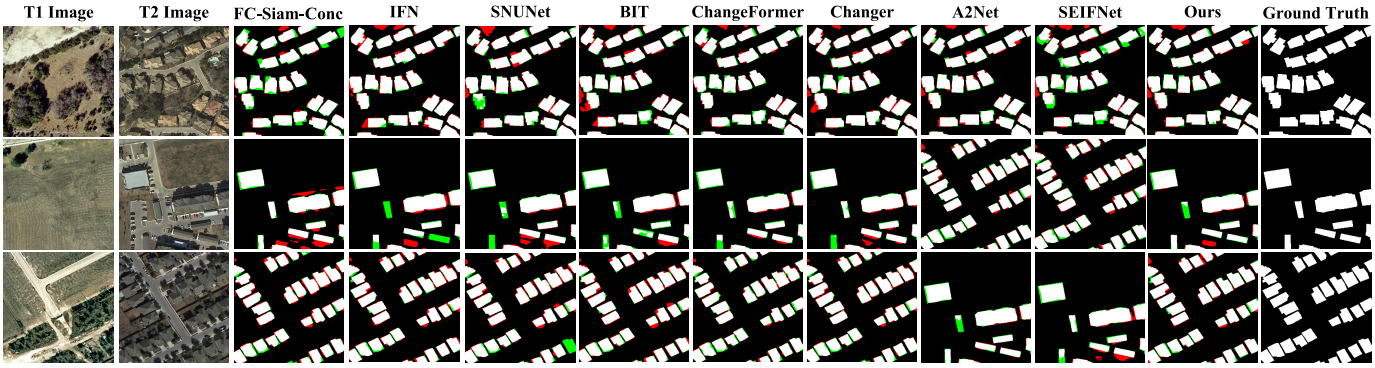


Fig. 7. Qualitative comparisons on the LEVIR-CD dataset. Different colors are used for a better view, i.e., white represents true positives, black represents true negatives, red represents false positives, and green represents false negatives. The less the green and red, the better the performance represented.

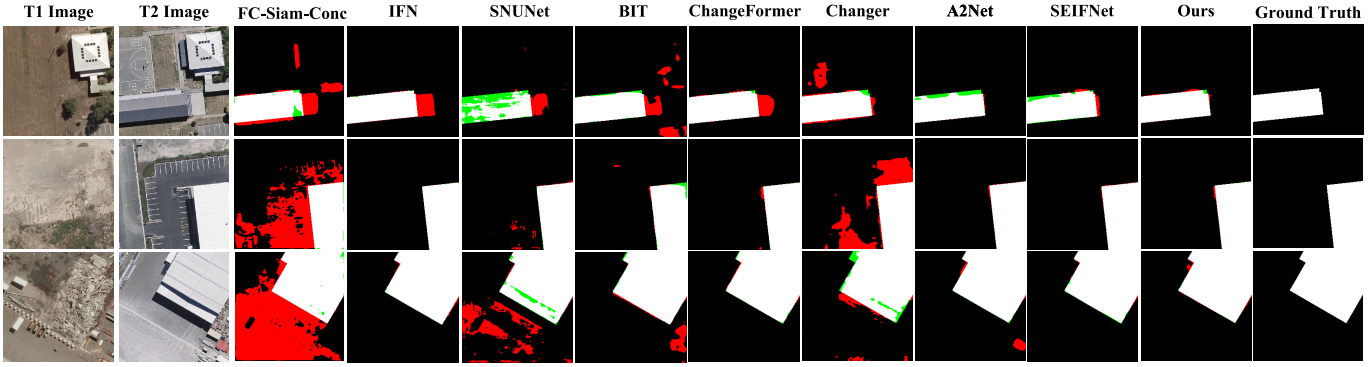


Fig. 8. Qualitative comparisons on the WHU-CD dataset. Different colors are used for a better view, i.e., white represents true positives, black represents true negatives, red represents false positives, and green represents false negatives. The less the green and red, the better the performance represented.

The PT-Former achieves the highest values in terms of recall and $F1$ in the LEVIR-CD dataset. A reasonable explanation is that PT-Former places more emphasis on position and temporal difference and is able to identify features that are ignored by other methods. Specifically, our method has a more effective ability to detect complex changes, which model the spatial and temporal information of changing features to accurately obtain the change areas.

Fig. 7 shows the CD results of all methods on different image pairs. In order to show the detection results more visually, we use different colors to indicate TP (white), TN (black), FP (red), and FN (green). PT-Former shows better performance in detecting shadow regions compared to other methods. Our method continues to outperform other methods in effectively handling the edges and structures of the focal content while successfully preventing structural loss when objects are modified. This indicates that PT-Former successfully avoids the pseudochanges and effectively combines the cross-temporal features by merging blocks.

3) Results on the WHU-CD Dataset: The quantitative comparison results are shown in Table I. The best values are marked in bold. PT-Former achieves the best results on all metrics except recall. Specifically, PT-Former is 0.05% and 0.40% higher than the second-ranked A2Net in precision/$F1$, respectively. Our methods aggregate the contextual information of the ground object by the fusion blocks, and the spatial relation is reconstructed under the guidance of bitemporal features. Quantitative evaluation results demonstrate the effectiveness and superiority of the PT-Former.

Fig. 8 shows the CD results of all methods on different image pairs. In order to show the detection results more visually, we use different colors to indicate TP (white), TN (black), FP (red), and FN (green). From the results of these challenging images, the proposed PT-Former is superior to other CD methods. Based on the detection from these difficult images, the PT-Former method outperforms other CD methods. When examining the incorrect and missed detection, PT-Former displays notably precise discriminant, even in the presence of occlusion and varying illumination. These results underscore the efficacy of PT-Former in emphasizing significant alterations while minimizing irrelevant disturbances.

B. Complexity Analysis

In order to ensure a fair comparison of model efficiency, we assess the complexity of PT-Former and other comparative methods based on the number of parameters (Params) and floating-point operations (FLOPs). Table I presents a detailed comparison of various networks in terms of Params and FLOPs. PT-Former demonstrates superior performance compared to other methods, with a balanced parameter count and FLOPs. Conversely, simpler architectures such as FC-EF, FC-Siam-Diff, and FC-Siam-Conv have fewer parameters and FLOPs but struggle to effectively manage spatial and temporal differences, leading to subpar performance. Our assessment highlights that PT-Former achieves a favorable balance between model complexity and performance, which makes it an option for bitemporal RS image CD tasks.

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TABLE II

ABLATION STUDY RESULTS OF THE PROPOSED PT-FORMER ON THE SVCD DATASET. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

PEM	TDPM	Pre-fusion block	Post-fusion block	Pre.	Rec.	F1	IoU	Kappa
				97.92	97.22	97.57	95.25	97.21
✓				98.06	97.24	97.65	95.40	97.30
	✓			98.26	97.15	97.70	95.50	97.36
✓		✓		98.27	97.31	97.79	95.67	97.53
✓			✓	98.16	**97.46**	97.81	95.71	97.46
✓		✓	✓	98.39	97.32	97.85	95.79	97.53
	✓	✓	✓	98.31	97.29	97.80	95.69	97.47
✓	✓	✓	✓	**98.46**	97.35	**97.91**	**95.90**	**97.60**

TABLE III

ABLATION STUDY RESULTS OF THE PROPOSED PT-FORMER ON THE LEVIR-CD DATASET. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

PEM	TDPM	Pre-fusion block	Post-fusion block	Pre.	Rec.	F1	IoU	Kappa
				92.96	90.57	91.75	84.76	91.31
✓				92.89	90.50	91.68	84.63	91.24
	✓			92.19	**91.51**	91.85	84.93	91.42
✓		✓		92.88	90.65	91.75	84.76	91.32
✓			✓	92.65	91.15	91.89	85.00	91.46
✓		✓	✓	92.95	91.00	91.96	85.12	91.54
	✓	✓	✓	93.28	90.60	91.92	85.05	91.49
✓	✓	✓	✓	**93.95**	91.04	**91.99**	**85.16**	**91.56**

TABLE IV

ABLATION STUDY RESULTS OF THE PROPOSED PT-FORMER ON THE WHU-CD DATASET. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

PEM	TDPM	Pre-fusion block	Post-fusion block	Pre.	Rec.	F1	IoU	Kappa
				96.05	91.63	93.79	88.30	93.54
✓				95.25	92.06	93.63	88.02	93.37
	✓			95.54	92.59	94.04	88.75	93.80
✓		✓		95.64	92.79	94.19	89.02	93.96
✓			✓	94.29	**92.88**	93.58	87.93	93.31
✓		✓	✓	95.79	92.81	94.28	89.18	94.05
	✓	✓	✓	95.79	92.81	94.28	89.18	94.05
✓	✓	✓	✓	**96.09**	92.63	**94.33**	**89.27**	**94.10**

C. Ablation Study

The ablation study results are presented in Tables II–IV, which is intended to confirm how well each component of the PT-Former performs, such as the PEM, TDPM, and two fusion blocks. Three different sets of ablation experiments were carried out on the SVCD, LEVIR-CD, and WHU-CD datasets to provide a thorough demonstration of each module's effectiveness.

1) Effectiveness of PEM: The PT-Former uses PEM to spatially model changes in dual-temporal features. As shown in Tables II–IV, after removing positional modeling, the metrics precision/recall/$F1$/IoU/Kappa decreased by 0.14%/0.06%/0.10%/0.10%/0.12% on the SVCD dataset, the metrics precision/recall/$F1$/IoU/Kappa decreased by 0.67%/0.44%/0.07%/0.01%/0.07% on the LEVIR-CD dataset, and the metrics precision/$F1$/IoU/Kappa decreased by 0.41%/0.04%/0.07%/0.04% on the WHU-CD dataset. These results indicate that PEM has successfully captured spatial changes in cross-temporal features.

2) Effectiveness of TDPM: The TDPM is designed to model distribution differences in cross-temporal features. To verify the effectiveness of TDPM, an ablation study is designed by removing TDPM. The results in Tables II–IV clearly demonstrate the advantage of TDPM in PT-Former, showing that the model's performance is enhanced after incorporating TDPM. After removing TDPM, the metrics precision/$F1$/IoU/Kappa decreased by 0.34%/0.13%/0.25%/0.15% on the SVCD dataset, the metrics recall/$F1$/IoU/Kappa decreased by 0.94%/0.08%/0.17%/0.11% on the LEVIR-CD dataset, and the metrics recall/$F1$/IoU/Kappa decreased by 0.96%/0.25%/0.45%/0.26%/ on the WHU-CD dataset. These results provide strong evidence that the proposed TDPM can model CTDs and adjust for these differences.

3) Effectiveness of Fusion Block: The fusion of bitemporal features in CD is essential as it helps combine different aspects of temporal information to form a comprehensive picture of changes. To carefully consider the incorporation of cross-temporal modeling through dual-stream features, we propose the fusion block to integrate contextual information of ground objects and reconstruct spatial relations guided by bitemporal features. To verify the effectiveness of fusion blocks, ablation studies are designed by removing fusion blocks. The results in Tables II–IV clearly demonstrated the necessity of dual-stream feature fusion before and after the transformer. The results show that after incorporating the pre-fusion block, the metrics precision/recall/$F1$/IoU/Kappa decreased by 0.21%/0.07%/0.14%/0.27%/0.23% on the SVCD dataset, the metrics recall/$F1$/IoU/Kappa decreased by 0.15%/0.07%/0.13%/0.08% on the LEVIR-CD dataset, and the metrics precision/recall/$F1$ decreased by 0.39%/0.73%/0.56%/ on the WHU-CD dataset. After incorporating the post-fusion block, the metrics precision/recall/$F1$/IoU/Kappa decreased by 0.10%/0.22%/0.16%/0.31%/0.16% on the SVCD dataset, the metrics recall/$F1$/IoU/Kappa decreased by 0.65%/0.21%/0.37%/0.22% on the LEVIR-CD dataset, and the metrics recall decreased by 0.82% on the WHU-CD dataset. After incorporating both two fusion blocks simultaneously, the model's performance achieves a greater improvement. These results provide strong evidence for the importance of cross-temporal feature fusion at multiple stages and demonstrate the effectiveness of the proposed fusion block.

D. Parameter Analysis

1) Discussion on the Number of the Transformer Cross-Encoder Heads: Table V presents the results of PT-Former with the different numbers of the transformer cross-encoder head. When the number of heads is 1 and 2, the metric is low with low computational complexity, but they cannot obtain the spatial information of the change areas from bitemporal features. For transformer cross-encoders with a large number of heads, the complexity increases, but more spatial information of change areas can be extracted. From Table V, the results show that the CD performance is improved with the increase in the number of the transformer cross-encoder heads. Finally, the number of heads in the cross-encoder of PT-Former is set as 8.

2) Discussion on the Patch Size of Patch Embedding: PT-Former uses patch embedding to reduce the size of the feature map, thus reducing the computational complexity of the model. However, the size of the patch directly affects the

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TABLE V
EFFECT OF THE NUMBER OF THE TRANSFORMER CROSS-ENCODER HEADS. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

Numbers	LEVIR-CD					WHU-CD				
	Precision	Recall	F1	IoU	Kappa	Precision	Recall	F1	IoU	Kappa
1	92.01	**91.63**	91.82	84.88	91.38	95.73	**92.97**	94.32	**89.28**	94.08
2	92.04	91.54	91.79	84.82	91.35	95.40	92.57	93.97	88.62	93.72
4	93.04	90.57	91.79	84.82	91.35	95.95	92.67	94.28	89.18	94.05
8	**93.95**	91.04	**91.99**	**85.16**	**91.56**	**96.09**	92.63	**94.33**	89.27	**94.10**

TABLE VI
EFFECT OF THE PATCH SIZE OF PATCH EMBEDDING. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

Sizes	LEVIR-CD					WHU-CD				
	Precision	Recall	F1	IoU	Kappa	Precision	Recall	F1	IoU	Kappa
4	92.69	91.20	91.94	85.09	91.51	94.90	**93.72**	94.31	89.23	94.07
8	**93.95**	91.04	**91.99**	**85.16**	**91.56**	**96.09**	92.63	**94.33**	**89.27**	**94.10**
16	92.37	**91.51**	91.94	85.08	91.51	95.52	92.91	94.20	89.04	93.96

TABLE VII
EFFECT OF THE BCE AND DICE LOSSES. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

Losses	LEVIR-CD					WHU-CD				
	Precision	Recall	F1	IoU	Kappa	Precision	Recall	F1	IoU	Kappa
BCE	92.75	90.23	91.47	84.29	91.02	95.41	91.36	93.34	87.52	93.07
Dice	92.83	89.99	91.38	84.14	90.93	95.62	92.27	93.92	88.53	93.67
BCE + Dice	**93.95**	**91.04**	**91.99**	**85.16**	**91.56**	**96.09**	**92.63**	**94.33**	**89.27**	**94.10**

TABLE VIII
EFFECT OF THE COEFFICIENT λ. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%)

λ	SVCD					LEVIR-CD					WHU-CD				
	Precision	Recall	F1	IoU	Kappa	Precision	Recall	F1	IoU	Kappa	Precision	Recall	F1	IoU	Kappa
0.6	98.43	97.25	97.83	95.76	97.51	93.09	90.72	91.89	84.99	91.46	96.16	91.97	94.02	88.71	93.78
0.8	98.43	97.32	97.87	95.84	97.56	92.95	91.19	91.85	84.93	91.42	95.48	**92.64**	94.04	88.74	93.79
1	98.45	**97.35**	97.90	95.88	97.59	**93.95**	**91.04**	**91.99**	**85.16**	**91.56**	**96.20**	92.52	94.32	89.25	94.09
1.2	98.45	97.33	97.89	95.86	97.57	93.19	90.64	91.90	85.01	91.47	96.09	92.63	**94.33**	**89.27**	**94.10**
1.4	**98.46**	**97.35**	**97.91**	**95.90**	**97.60**	92.68	91.07	91.87	84.96	91.43	95.40	92.64	94.00	88.68	93.76

amount of semantic information in the features. As shown in Table VI, we experimented with the impact of patches of different sizes on the performance of CD. The experimental results indicate that when the patch size is 4, the model's performance is not optimal. It reaches the best performance when the patch size is 8, but the performance decreases when the patch size is 16. This suggests that a patch size of 8 contains the most appropriate semantic information.

3) Discussion on the Hybrid Loss: This article introduces a hybrid loss function that combines BCE loss with Dice loss for CD tasks. The hybrid loss is designed to effectively consider both pixel-level similarity and object-level overlap between predicted and actual change maps. To evaluate the effectiveness of the hybrid loss, we compare models trained with only BCE loss or Dice loss, as detailed in Table VII. The results demonstrate that utilizing Dice loss yields better results than using BCE loss alone. Moreover, models trained solely with BCE or Dice loss demonstrate lower performance in CD compared to models incorporating the hybrid loss. These results suggest that the hybrid loss, which integrates BCE and Dice losses, is advantageous for CD tasks. By combining pixel- and object-level information, the hybrid loss enhances the model's ability to accurately identify changes in the input data. Therefore, we choose to implement the hybrid loss after each decoder stage in our experiments.

4) Discussion on the Loss Function Coefficient: This article introduces a coefficient λ for balancing BCE loss and Dice loss in the hybrid loss function. To verify the effect of different values of λ on the experimental results, we select five values of λ and conduct experiments on three datasets. The experimental results are shown in Table VIII. On the SVCD dataset, the performances of PT-Former are almost identical. When λ is 1.4, the metrics precision/$F1$/IoU/Kappa improved by 0.01%/0.01%/0.02%/0.01%, respectively. Therefore, on the SVCD dataset, we set λ to 1.4. The results indicate that the model achieves the best performance when λ is set to 1 in the LEVIR-CD dataset, which suggests that BCE loss and Dice loss have the same importance. Therefore, on the LEVIR-CD dataset, λ is set to 1. On the WHU-CD dataset, the performances of PT-Former are almost identical when λ is

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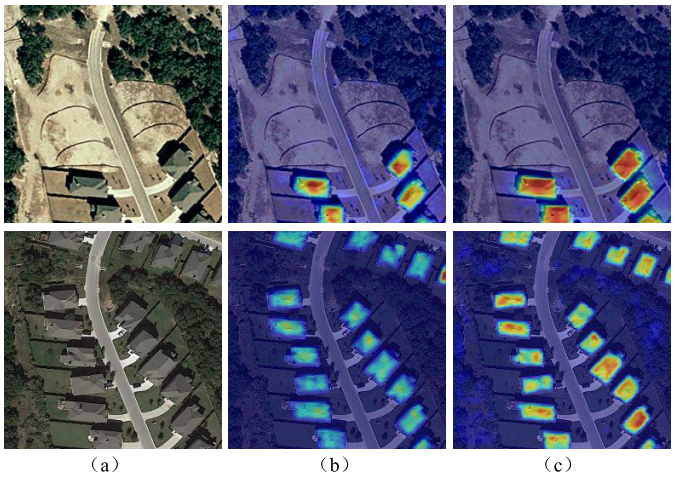


Fig. 9. Feature visualization on the LEVIR-CD dataset. (a) Input images. (b) Features w/o PEM. (c) Features w/ PEM.

set to 1 and 1.2. When λ is 1, the model achieves the highest precision of 96.20%. However, when λ is 1.2, the metrics $F1$/IoU/Kappa improved by 0.01%/0.02%/0.01%, respectively. Therefore, on the WHU-CD dataset, we set λ to 1.2.

E. Feature Visualization

In order to better understand the feature learning from the PEM in the proposed PT-Former, we take a pair of images from the LEVIR-CD dataset as an example and use Grad-CAM to visualize these features in Fig. 9. In the feature visualization maps, red indicates the areas that the model pays more attention to, while blue represents the areas that the model pays less attention to. From the visualizations, it can be seen that the model's perception of ground objects is unclear without PEM. After PEM is added, the model's detection of ground objects is enhanced, and the description of object edges is more accurate. This demonstrates that the proposed PEM can model spatial changes and enhance the model's capability for CD.

VI. Conclusion

In this article, we proposed a PT-Former that focuses on modeling position and temporal relations within bitemporal images. The PT-Former involves utilizing a Siamese network integrated with a PEM to function for identifying changes in specific areas. In addition, a TDPM has been developed to analyze temporal shifts and enhance the ability to perceive differences across different temporal. Furthermore, the fusion blocks are employed to consolidate contextual information related to the ground objects, while the spatial relations are restructured based on the bitemporal features. Experiments on three benchmark datasets verify the effectiveness of the proposed PT-Former. Moreover, the main limitation of the proposed PT-Former under consideration is its high complexity. In the future, our attention will be directed toward designing a lightweight CD model to reduce computational complexity.

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